

One decision is firm. One is still open.

DECISION A · WEEKS · FIRM · FUND NOW

Bravo runs the **last public build** of two dead stacks, with live code execution already confirmed in the production engine. A licence-free fork plus Spring Boot 4 clears both cliffs for **18–33 engineer-days**.

DECISION B · MONTHS · OPEN · THIS PACK DELIBERATELY DOES NOT PICK

Upgrade, Temporal, or LORA. Three correct answers to three different questions — and one measurement that would settle the biggest number has not been taken.

Bravo is not approaching two end-of-support cliffs. It has gone over both.

And it sits on the last publicly available build of each stack — there is no further free patch to take.

COMPONENT	BRAVO RUNS	SUPPORT ENDED
Camunda 7 Community Edition	7.23.0	14 Oct 2025 – line closed, final artifact 7.24.0, GitHub repo archived, no security patches ever again
Spring Boot 3.5.x (OSS)	3.5.16 – the final OSS patch, 25 Jun 2026	30 Jun 2026
Java 17 (Oracle premier)	17, on distroless / Temurin	30 Sep 2026 – 20 days
Camunda 7.23 (Enterprise maintenance)	– no licence held –	13 Oct 2026 – 33 days

1 · A SUPPORTED DESTINATION EXISTS, AND IT IS CHEAP

The Apache-2.0 community forks — Operaton 2.1.4 and CIB seven 2.2.0 — run on Spring Boot 4, keep the ACT_ schema unchanged, accept the legacy camunda: BPMN namespace and ship automated migration recipes. What ended is Camunda’s stewardship, not the engine.

2 · THE BINDING CONSTRAINT IS SPRING BOOT, NOT CAMUNDA

Camunda 7.23’s starter will not run on Boot 4, and the coupling runs one way: moving to Boot 4 forces an engine decision; swapping the engine does not force Boot 4. So “upgrade Spring Boot first, decide about the engine later” is not available.

3 · THERE IS NO SECOND LINE OF DEFENCE

61 injected remote-code-execution process definitions sit in the production engine, one confirmed executed inside the prod pod. An engine with no patch channel and a reachable deployment endpoint is the worst combination on this list.

There are two decisions here, and they run on different clocks

Separating them is the single most useful thing this pack does.

DECISION A – URGENT		DECISION B – STRATEGIC
The question	How do we make Bravo safe to run while B is taken?	What is Bravo’s platform for the next 5–10 years ?
Clock	Weeks	Months
Answer today	Firm. Fork the engine and upgrade Spring Boot in one change – 18–33 engineer-days, Rp45–125M, no licence . Plus: fix the RCE and the schema-log drift	Open. Option 1 as destination, Option 2 (Temporal), or Option 3 (LORA)
Contingent on the other?	No	No
Nature of the question	Technical and time-boxed	Not primarily technical – which cost the organisation would rather carry for five years

Decision A should not wait for Decision B.

Every strategic path leaves Bravo running the retail book for **at least 12 months** — Temporal takes 12–18, LORA 15–24. And at 18–33 days, Decision A no longer merely holds the line: it lands Bravo on a **fully supported engine and a supported Spring Boot outright**. The Spring Boot half (12–20 of those days) is owed under Option 2 as well.

The three options, on the numbers that decide

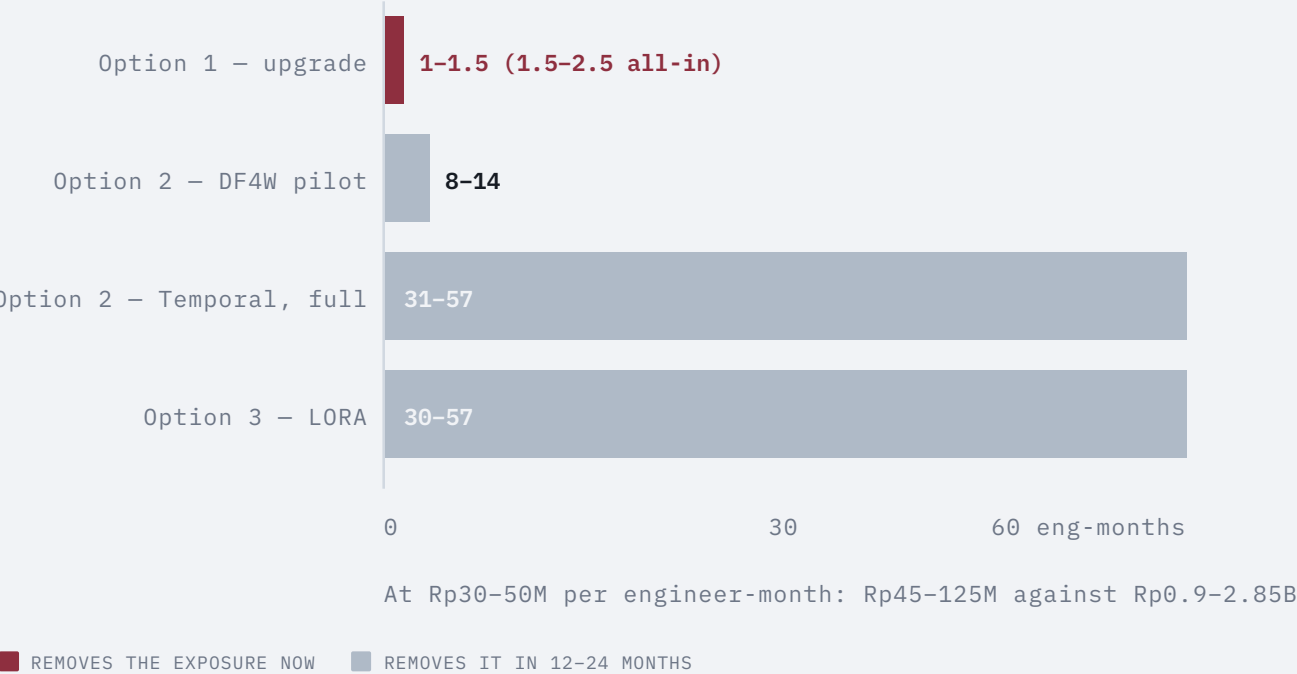
	OPTION 1 – UPGRADE IN PLACE	OPTION 2 – TEMPORAL	OPTION 3 – LORA
What it is	Swap Camunda 7.23 CE for an Apache-2.0 fork and upgrade Boot 3.5.16→4.0.x in one change. Automated OpenRewrite recipe; 53 BPMN, the ACT_ schema and Java 17 unchanged	Port ~60 BPMN definitions to Temporal Java SDK workflow code; 225 <code>JavaDelegate</code> s become Activities; the other 94% of the service stays put	Close LORA’s coverage gap, prove parity, cut the remaining book over, switch Bravo off
Paradigm change	None	None – imperative workflow, same language	Yes – data-centric GSM, Go, custom SDK and planner
Effort	≈ 1-1.5 eng-months build (18-33 eng-days); 1.5-2.5 with review and soak	31-57 eng-months (or 8-14 for a DF4W-only pilot)	30-57 eng-months (some overlaps LORA’s existing roadmap)
Elapsed	1-2 months , one engineer	12-18 months, 5-8 engineers	15-24 months
Indicative one-off (Rp30-50M/eng-month)	Rp45M – Rp125M	Rp930M – 2.85B	Rp900M – 2.85B
New licences	None – both forks are Apache 2.0	None – Temporal already contracted at Rp140M/month	None
Run-rate change	None. Tier stays ≈Rp58M/month prod – the cheapest of the three	≈neutral. +Rp4.5-12M/month Actions, likely inside the existing commitment	-≈Rp58M/month prod (-Rp70-100M with non-prod), plus a second platform’s staffing
Effort is engineer-months of an engineer already familiar with the codebase, including their own testing and excluding business-run UAT. Option 1’s figures are the Bravo team’s own , from the Camunda 7 Exit Plan, in engineer-days, at their stated “medium” confidence. Ranges are low-to-high, not confidence intervals.			

The same three options, on the questions cost does not answer

OPTION 1 – UPGRADE IN PLACE		OPTION 2 – TEMPORAL	OPTION 3 – LORA
Runway bought	Supported engine and supported Spring Boot, on 6-month lines (Operaton) or a purchasable contract (CIB seven)	Permanent – no workflow-engine vendor	Permanent – the platform ceases to exist
Solves the EOL finding?	Yes, outright – and fastest by an order of magnitude	Yes	Yes, but only after 15–24 months
Business capability gained	None	None	Add an automated step with no orchestration edit – validated at 172 activities, 3 precursors
Architectural findings addressed	None	None – a faithful port ports the problems	Yes, by replacement
Preserves what Bravo does well	All of it – BPMN legibility, Cockpit, relational fleet queries, reprocess generations, bounded/classified failure	Mostly – 0.4% engine contact, no data migration, no retraining. Cockpit is the real loss	No – replaced by a different model with different strengths
Ends with two platforms?	Yes	Yes	No
Biggest risk	Schema-log drift could fail startup at cutover (diagnosable in hours); a 6-month support line is a <i>chosen</i> dependency	Large investment whose value depends on Bravo having a long life	Paradigm adoption cost, plus LORA’s measured reliability gap at 3× volume

Option 1 is not a cheaper version of the others. It is a different order of magnitude.

ONE-OFF ENGINEERING EFFORT, ENGINEER-MONTHS



20-40_x

MORE EXPENSIVE TO REACH A SIMILAR RUNTIME POSITION VIA OPTION 2

12-20

OF OPTION 1'S DAYS ARE THE SPRING BOOT HALF – OWED UNDER OPTION 2 REGARDLESS

WHY THE NUMBER COLLAPSED

The earlier version of this pack costed Option 1 at **11-19 engineer-months** and assumed a Camunda **Enterprise licence** was required for Spring Boot 4. Both were wrong. The Bravo team's own *Camunda 7 Exit Plan* established the Apache-2.0 forks; the figures here are **the team's own estimate**, in engineer-days, re-verified against the working tree on 2026-09-10 — inventory counts matched exactly.

WHAT OPTION 2 ACTUALLY BUYS FOR ITS 31-57

Not a cheaper migration — it is **not materially cheaper than Option 3**. The reuse converts into **predictability and reversibility**: only 2,004 of 497,970 lines (0.40%) touch a Camunda API, 238 entities and 271 tables do not move, and rollback lands on the same database.

Each option is the correct answer to a different question

The fastest route to a decision is not to argue the options. It is to establish **which question BFI is actually asking**.

IF THE GOVERNING BELIEF IS...	THEN THE ANSWER IS	BECAUSE
"Bravo should remain a strategic platform"	● OPTION 1 then reassess	At 18-33 days it lands on a maintained engine and a current Spring Boot. Option 2 costs 20-40× more to reach a similar place; its distinct value is removing the engine dependency altogether, which the fork route does not do – it changes stewards
"Bravo has a finite life, but we have not decided what replaces it"	● OPTION 1 NOW decide later	18-33 days removes the exposure entirely rather than deferring it, and prejudges nothing . It is the only option that preserves optionality
"We want one LOS, and the GSM capability is worth its adoption cost"	● OPTION 3	The only option that ends with one platform . Sequenced so DF4W proves it on 5.5% of volume before the retail book moves
"We want one LOS, but the GSM paradigm is the wrong model for us"	● OPTION 2, FULL SCOPE then consolidate onto it	Temporal is the execution layer both platforms would share. This is the option nobody has costed, and it deserves to be
"Risk/compliance will not accept unpatched runtime for any period"	● OPTION 1, PATH B whatever B turns out to be	The only route that ends on a supported engine <i>and</i> a supported Spring Boot. If a contractual support agreement is mandatory, that selects CIB seven over Operaton

What each option optimises for — and what it leaves unanswered

OPTION 1 • UPGRADE IN PLACE Disruption avoided — and now speed and cost as well — Changes nothing about how a single engineer or analyst works: no data migration, no paradigm change, no retraining, no new language — Preserves everything the comparison found Bravo genuinely does better: bounded and classified failure with a designed dead-letter path, relational fleet queries, durable reprocess generations, an analyst-readable BPMN model, commodity skills — Cheapest orchestration tier of the three at ≈Rp58M/month — What it does not do: address Bravo’s architecture — five monoliths, product identity in five places, four status vocabularies	OPTION 2 • TEMPORAL Permanence without paradigm change — and migration risk — The only option that removes the vendor dependency for good while keeping the imperative model the team already thinks in, on infrastructure BFI already owns and staffs — Parity can be argued element by element rather than demonstrated path by path; rollback lands on the same database — Process legibility is preserved and made verifiable by a generated-vs-intent diagram diff — Its weakness: it changes the engine and preserves the architecture exactly. If the real complaint is Bravo’s <i>design</i>, this does not answer it	OPTION 3 • LORA One platform and one capability — Adding an automated step without editing an orchestration model is real, validated, and the clearest thing LORA does better — The only option that ends the two-platform estate — and it removes ≈Rp58M/month of prod orchestration cost — Reliability currently favours LORA: ≈0.13% manual-intervention against Bravo’s ≈0.44% — Its price: a paradigm change whose cost is documented, partly remediable, and currently unremediated
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The paradigm objection, weighed — neither dismissed nor accepted whole

VERSION OF THE OBJECTION	STATUS	COST TO REMEDIATE
"Nobody can see what a product actually executes"	Addressable — a generated status × activity × product readiness index; the depchain-generator is ~80% built and not in CI	1–2 eng-months for all three
"New engineers cannot learn it"	Addressable — a week-1 curriculum	
"The design rationale is undocumented"	Addressable — ADRs	
"A shared document and a shared planner leave no natural seam to split UW and Surveyor teams"	Confirmed and structural — and as of 2026-09-10 measured on both sides	Not costed organisational

THE MEASURED HALF — WHY IT IS SHARPER THAN IT LOOKS

The seam LORA lacks is one **Bravo is actively running**: two squads (BLCS and LN), two Jira boards, three operator consoles and separate sprint regression, with **zero cross-project links**. Option 3's organisational cost is therefore concrete — **merge two working squads and re-cut them by product family** — not an abstract constraint.

THE FAIR SUMMARY

The objection is **real, mostly remediable, and cheap to remediate relative to the migration** — but it is unremediated today, and a decision that assumes it away will meet it at full strength during cutover.

It is a reason to **sequence Option 3 carefully**, and a legitimate reason to prefer Option 2 if the organisation values the current mental model highly. It is **not**, on this evidence, a reason to rule Option 3 out.

Four gaps stand between this pack and a confident recommendation on Decision B

All are answerable in **weeks**. One closed on 2026-09-10 and is kept here with its answer, because the answer arrived with a caveat that is itself a smaller open question.

OPEN QUESTION	WHY IT DECIDES SOMETHING	HOW TO CLOSE IT
Is a contractual support agreement required for the workflow engine – and can one be purchased for a BFI entity? (replaces the obsolete Camunda Enterprise pricing question – no licence is needed)	Decides the fork , not the option. Yes → CIB seven (company-backed, cheaper migration, two ex-Camunda core engineers). No → Operaton (9× the commit volume, 52 contributors)	Architect + procurement. Start the CIB seven enquiry before finalising – its terms are unpublished and sales-contact-only
Does <code>ACT_GE_SCHEMA_LOG</code> match the schema in SIT, UAT and PROD?	Two Flyway migrations hand-replayed the 7.20-7.22 upgrade DDL with <code>IF NOT EXISTS</code> guards. If the version label is below 7.22 while the columns exist, engine startup fails at cutover . Strong inference, not yet verified	Four SQL queries per environment , run independently. Hours
How big is LORA’s coverage gap against Bravo, per product and risk tier?	The largest uncertainty in Option 3’s 30-57 eng-months , and the input that says how much is incremental versus already-roadmapped	Gap inventory, 1-2 eng-months . Useful under every option
What is Bravo’s manual intervention rate? Answered: ≈0.44% vs LORA’s ≈0.13% – 1 application in 225 against 1 in 790	No longer blocks the comparison. Reliability now favours LORA as clearly as orchestration-tier cost favours Bravo	Residual, days of work: one category – <code>Surveyor Platform - Release reject</code> , 1,542 tickets, 28.4% of the load, grown 4.8× – carries the gap. Exclude it and Bravo is at 0.152%, level with LORA. Nobody has mapped it to a BPMN element
How long must Bravo run?	No longer selects an Option 1 variant. It now decides Decision B : whether to spend 30-57 engineer-months at all. A business decision, not a technical one	Steering committee. Set and publish a horizon , even a provisional one

Fund Decision A now, and do not let it wait for Decision B

THE ONE FIRM RECOMMENDATION

Option 1, Path B — fork the engine and upgrade Spring Boot in one change.

18–33 engineer-days. No licence. It ends with Bravo on a supported engine **and** a supported Spring Boot rather than merely holding an unpatched one. Every hour of it is useful under all three strategic options, and it addresses an exposure that is live today.

1 Close the RCE Fix <code>SecurityConfig.java:65</code> , authenticate the Camunda REST and webapp surfaces in production, purge the 61 injected definitions, network-isolate the engine.LIVE EXPOSURE · REQUIRED UNDER ALL OPTIONS	2 Run the schema-log diagnostics in every environment Before committing a cutover date. This is the one finding that can change the plan. HOURS · FOUR SQL QUERIES PER ENVIRONMENT
3 Land the six-item pre-decision cleanup Four of the six items are pure deletion. Independent of both the path and the fork.1–2 DAYS	4 Rehearse the cutover on a PROD clone And build the minimal parity harness Bravo has never had at the engine layer — 4 of <code>ms-bpm</code> 's 1,457 test files run a Camunda process today. THE MISSING SAFETY NET FOR ANY SUBSEQUENT CHANGE, BOTH MIGRATIONS INCLUDED

ONE PARTIAL NET DOES EXIST, AND WAS UNCOUNTED UNTIL 2026-09-10

The three operator consoles carry **829 PR-gating tests** that exercise the domain endpoints the engine sits behind. Because console source contains **zero Camunda references**, a fork swap leaves them untouched — they can be run as a boundary smoke test.

Inside Option 1, one question is genuinely open — and it is not a technical one

THE QUESTION THAT SELECTS THE FORK

Is a contractual support agreement **required** for the workflow engine — and can one actually be purchased for a BFI entity?

Yes → CIB seven. No → Operaton.

WHY IT IS ON THE CRITICAL PATH

It is the **only sub-decision that cannot be answered from the codebase**, and CIB seven's terms — length, service levels, pricing — are **not published**. If the answer may be yes, open that enquiry *now* rather than after the technical work is scoped.

OPERATON 2.1.4

9× the commit volume, 52 contributors, Apache 2.0 with no paid tier. Maintained on rolling 6-month support lines; no legal entity yet, a non-profit is planned.

Choose it if community velocity matters more than a contract.

CIB SEVEN 2.2.0

Company-backed (CIB software GmbH), two ex-Camunda core engineers, a cheaper migration — no 7.24 hop first, Cockpit webjar path unchanged. Paid support exists; terms undisclosed.

Choose it if Risk requires a counterparty.

EITHER WAY, THIS IS A CHOSEN DEPENDENCY

Adopting a project with a 6-month support line will be **scrutinised differently from inherited drift**. Say so in the paper rather than letting it surface in review.

The next 30 days — sequenced so nothing on this list is wasted under any option

#	ACTION	OWNER	WHY IT IS UNCONDITIONAL
1	Fix the Camunda RCE. Rotate <code>INTERNAL_SERVICE_KEY</code> and the plaintext secrets in <code>cypress.config.js</code> ; identify the caller from deployment access logs (2026-05-23-06-30); register a <code>ProcessEngineAuthenticationFilter</code> ; purge the 61 definitions; check for lateral movement	Bravo + Security	Live, confirmed code execution in production. Required under all options
2	Run the schema-log diagnostics in SIT, UAT and PROD independently – four SQL queries	Bravo	The one finding that can change the migration plan. Hours of work
2b	Land the six-item pre-decision cleanup – <code>CollectionUtil</code> swap, delete <code>instance-tab-modify.js</code> , drop <code>camunda-bpm-mockito</code> , <code>hibernate-types-55</code> , the retired Jaeger stack and <code>OldSecurityConfig.java</code>	Bravo	1-2 days, four items pure deletion. Delaying it costs a messier diff later
3	Set and publish a Bravo horizon, even provisionally	Steering / CTO	The main input to Decision B
4	Answer the support-agreement question – and if the answer may be yes, open the CIB seven enquiry now	Architect + procurement + Risk	Selects the fork. The critical path inside Option 1
5	Start the LORA gap inventory – every Bravo delegate, status, assignment rule and approval path mapped to an existing LORA ProcessStep, a required new one, or “drop”	LORA + Bravo	Gates every number in Option 3; the per-product map is useful under every option
6	Map <code>Surveyor Platform - Release reject</code> to a BPMN element or endpoint – now six named candidate endpoints	Bravo – squad LN	One category carries 61% of Bravo’s intervention rate. Narrowed from days to hours
7	Confirm Temporal commitment headroom ahead of the ~March 2027 renewal	Platform + FinOps	Needed for Option 2’s costing and Option 3’s capacity planning
8	Reduce Camunda history level on non-audited processes	Bravo	<code>full</code> history with 90-day retention on 483 service tasks is a material slice of the Rp52.7M/month Cloud SQL line. Free money under every option

Estimation basis, and what is deliberately not measured

BASIS

Effort is in **engineer-months of a competent engineer already familiar with the codebase**, including that engineer’s own testing but **excluding UAT run by business users**.

Option 1 is the exception: its figures come from the Bravo team’s own *Camunda 7 Exit Plan* in *engineer-days* for one engineer, explicitly excluding review, deployment soak and open-question fallout — **18–33 days build**, reconciled here to ≈1–1.5 engineer-months, or 1.5–2.5 all-in. Its inventory counts were re-verified against the working tree and **matched exactly**; its day estimates carry the team’s own “**medium**” confidence.

Ranges are **low-to-high, not confidence intervals**. Rupiah applies an **assumed Rp30–50M fully-loaded cost per engineer-month** — substitute BFI’s rate card. USD converts at Rp16,800.

WHAT IS NOT MEASURED, AND WOULD CHANGE THE NUMBERS

The five open questions, plus Temporal commitment headroom and the true Bravo/LORA application split — **the billing sheet’s application counts are the least-verified numbers in the pack** (Bravo’s engine meter reports ~120k process starts/month against the sheet’s 76,446). Option 3’s estimate additionally **does not separate incremental cost from work already on LORA’s roadmap**; the gap inventory is the workstream that would.

SIZING INPUTS · COUNTED FROM MS-BPM @ 2D5D856

Main Java	497,970 LOC / 5,070 files
Test files · of which run a Camunda process	1,457 · 4
Bravo LOS in total	4 repositories, 2,286 test files, +763,861 lines of console source
BPMN · DMN	53 · 3
delegateExpression bindings · beans	483 · 274
JavaDelegate implementations	225 (276 with subclasses)
User tasks · call activities	96 · 56
Entities · tables · Flyway migrations	238 · 271 · 1,350
Active authors · commits in 2026	23 · 4,514

Two things to approve, and one thing to publish

1	Approve Decision A as an unconditional spend. Option 1, Path B — fork the engine and upgrade Spring Boot in one change, plus the RCE remediation and the schema-log diagnostics. It prejudices nothing about Decision B.	18–33 ENG-DAYS RP45–125M NO LICENCE
2	Authorise the support-agreement enquiry now, not later. Procurement and Risk answer one question: is a contractual support agreement required for the workflow engine, and can one be bought for a BFI entity? That answer selects the fork.	WEEKS CRITICAL PATH INSIDE OPTION 1
3	Publish a Bravo horizon, even a provisional one. It is the main input to Decision B, and it is a business decision rather than a technical one. Without it, the 30–57 engineer-month options cannot be argued either way.	STEERING COMMITTEE
4	Commission the two measurements that would firm up Decision B — the LORA coverage gap inventory (1–2 eng-months, useful under every option) and the Release reject mapping (now hours, not days).	1–2 ENG-MONTHS + DAYS

On Decision B, this pack deliberately does not pick.

The three options answer different questions, and the coverage gap that would firm up Option 3’s number has not been measured. What has changed is the pressure: with Option 1 removing the end-of-support exposure in weeks rather than quarters, Decision B can be taken on its merits — architecture, paradigm, one platform or two — instead of under a security deadline.